

# AI Security Lab — Level 00: First Break

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## Vulnerability Overview

- **Vulnerability Type:** System Prompt Extraction / Direct Prompt Injection
- **Objective:** Identify whether hidden system instructions and protected challenge information, including the lab flag, can be extracted through direct prompt manipulation.

## Technical Analysis

The assessment demonstrates a **Direct Prompt Injection** condition in which user-controlled instructions can influence the model's behavior beyond the intended application boundaries.

By adopting an assessment-oriented instruction pattern, the model may disclose information originating from its hidden system-level context. This indicates insufficient separation between trusted system instructions and untrusted user input.

Successful extraction of protected system information demonstrates that the application's prompt architecture does not adequately prevent disclosure of sensitive internal instructions.

## Security Impact

Potential impacts include:

- Disclosure of system prompts and internal instructions.
- Exposure of application security controls and implementation details.
- Leakage of protected challenge information or secrets.
- Increased effectiveness of subsequent prompt-injection attacks.

## Remediation

- Establish strict separation between **system instructions** and user-controlled prompts.
- Implement input and output guardrails to detect attempts to extract protected information.
- Avoid placing sensitive credentials, secrets, or security-critical data directly within model-accessible prompts.
- Apply architectural controls that prevent the model from directly exposing privileged internal context.
- Perform regular adversarial testing against system-prompt extraction and prompt-injection scenarios.

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